Estimating the preference for safety features in cars

Model: Multinomial logit and Mixed logit (discrete choice)

The Analytics Edge: Companies carry out conjoint studies with customers to understand how they tradeoff among the attributes that make up a product or a service. Using data of preferences of customers for new safety features in vehicles and then building models of discrete choice, the company can obtain estimates on how customers value attributes and bundles of attributes. This provides important information that can be used to infer the effect of introducing new products in the market, pricing these products and designing a product.

The models incorporate the comparison of attributes across alternatives and can also be used to capture heterogeneity in preferences in the choice modelling process.

Stated and revealed preferences

Stated choice: In stated choice questionnaires or surveys, the respondent is asked about his or her preferences over a series of (hypothetical) choices. The advantage of such experiments is that it allows one to study choice situations over a range of variations in the attributes and even over a range of choices that might not exist within the current outcomes. It is often relatively easy to conduct such stated choice experiments. On the other hand, since it is only a stated preference, it might not reflect actual choice behavior and users might for example overstate their valuation of a product or service.

Revealed choices: In revealed preference data, we try to infer the preferences/choice behavior from the actual choices that people made. For example, we might look at the products that customers bought from an online retail site when they are shown an assortment to understand their true choice behavior. This requires revealed preference data to be available. On the other hand in such an approach, it is harder to understand preferences over new alternatives where actual choices have as yet not been revealed.

Summary

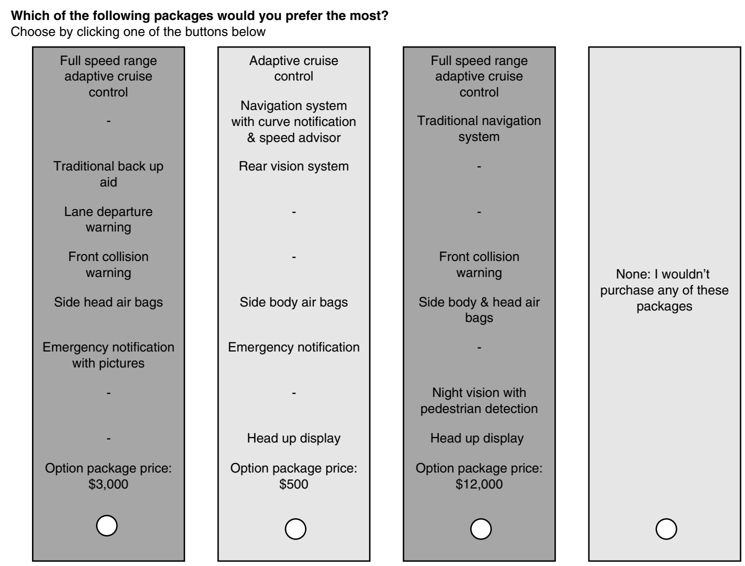
Data: Source is stated preference data on new safety features of cars (obtained from a consulting project).

Model: Multinomial logit and mixed logit are used to predict preferences over these features.

Decision: The model identifies the tradeoff among the attributes of products which in turn helps with pricing and product design decisions.

Value: Understanding preferences of customers is key to much of what most companies do.

Figure 4 A Sample Choice Task



and MEM. We compare the three models, both with and without heterogeneity. We estimated the parameters for MNL, NestL, and MEM by maximizing the log-likelihood function. For each respondent $i \in \mathcal{I} = \{1, \ldots, 500\}$, we use the first 12 of the 19 choice tasks performed by the consumer to estimate the parameters of the models. Thus, each consumer i performs $j \in \mathcal{J} = \{1, \ldots, 12\}$ choice tasks in the calibration step. This corresponds to a total of 6,000 in-sample data points (consumer-choice task pairs). The set of available

Table 1 Attribute and Level Codes

Serial no.	Attribute name	Attribute code	No. of levels	Level codes
1	Cruise control	CC	3	CC1, CC2, CC3
2	Go notifier	GN	2	GN1,GN2
3	Navigation system	NS	5	NS1, NS2, NS3, NS4, NS5
4	Backup aids	BU	6	BU1, BU2, BU3, BU4, BU5, BU6
5	Front park assist	FA	2	FA1, FA2
6	Lane departure	LD	3	LD1, LD2, LD3
7	Blind zone alert	BZ	3	BZ1, BZ2, BZ3
8	Front collision warning	FC	2	FC1, FC2
9	Front collision protection	FP	4	FP1, FP2, FP3, FP4
10	Rear collision protection	RP	2	RP1, RP2
11	Parallel park aids	PP	3	PP1, PP2, PP3
12	Knee air bags	KA	2	KA1, KA2
13	Side air bags	SC	4	SC1, SC2, SC3, SC4
14	Emergency notification	TS	3	TS1, TS2, TS3
15	Night vision system	NV	3	NV1, NV2, NV3
16	Driver assisted adjustments	MA	4	MA1, MA2, MA3, MA4
17	Low speed braking assist	LB	4	LB1, LB2, LB3, LB4
18	Adaptive front lighting	AF	3	AF1, AF2, AF3
19	Head up display	HU	2	HU1, HU2
20	Price	Price	11	\$500, 1,000, 1,500, 2,000, 2,500, 3,000, 4,000, 5,000, 7,500, 10,000, 12,000

This is taken from the paper On Theoretical and Empirical Aspects of Marginal Distribution Choice Models, Management Science, Special Issue on Business Analytics, V. K. Mishra, K. Natarajan, D. Padmanabhan, C-P. Teo and X, Li, 2014